Muhammad Muizz Zafar

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Software Engineer | ETL Pipelines | Data Platforms | Cloud Infrastructure

Skills

Programming Languages:	Java, Python, TypeScript, GoLang, SQL, JavaScript, Kotlin, C#, C++, HTML/CSS, PHP
Tools/Frameworks:	AWS, AWS CDK, CloudFormation, Docker, Splunk, CloudWatch, Athena, Git, Linux

Education

University of Ottawa — Ottawa, ON
Bachelor of Applied Science, Software Engineering *GPA: 3.7/4.0* — Sept 2020 - March 2025

Work experience

ETL Verification Developer — May 2024 - Present
Elections Canada — Ottawa, ON

- Built automated **ETL testing framework** using **Python** and **Azure Devops Pipelines** to ensure efficient data movement validation across **20M+ records**, reducing manual validation time.
- Used **Apache Spark** to profile, transform, and validate large datasets (up to **5GB**) in staging and production environments, improving anomaly detection and bulk data checks.
- Developed internal **data analytics dashboards** using **Python**, **Streamlit**, and **Plotly Dash** to surface **pipeline health**, operational metrics, and data quality trends for senior management.
- Coordinated version-controlled database deployments and **release processes**, enhancing traceability and reliability of **production data infrastructure**.
- Collaborated within an **Agile cross-functional team** with stakeholders to improve **distributed data workflows** and production reliability.

Software Developer — September 2024 - December 2025
Kinaxis Inc. — Ottawa, ON

- Developed an automation framework to convert **35,000 legacy KUnit tests** into **JUnit** in **Java**, improving test reliability and coverage for backend algorithms.
- Designed **automated SQL queries** to detect **50+ known behavioral differences** between legacy and modular codebases, streamlining code comparison and testing.
- Developed **Java services** for scope control to detect incompatibilities in the **Rapid Response product**, reducing **L3 escalations** from customers by improving product reliability.
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- Used **JavaScript** for building automated workflows to visualize differences in the product, enhancing communication across product teams.

Full Stack Developer Intern — September 2023 - April 2024
University of Ottawa — Ottawa, ON

- Redesigned the University of Ottawa's **emergency alert website** using **Laravel** and **PHP**, improving user experience for **20,000+ students daily**.
- Enhanced **Drupal**-based features for the **main website**, benefiting **50,000+ students**.
- Optimized **Java REST APIs**, reducing **SQL query execution time** for event data lookup by **15 seconds**.

Cloud Developer Intern — May 2023 - August 2023
Trend Micro Inc. — Ottawa, ON

- Developed serverless **monitoring workflows** using **AWS Lambda**, **S3**, and **CloudWatch** to analyze **50,000+ cloud artifacts** daily, supporting automated **vulnerability scanning pipelines**.
- Defined cloud infrastructure using **Golang**, **AWS CDK**, and **CloudFormation**, provisioning **50+ AWS resources**.
- Implemented concurrent processing logic in **GoLang** using goroutines, improving throughput for distributed cloud data processing workloads.
- Integrated **vulnerability scanning tools** like **Anchore** and **Grype** with AWS services for automated security checks.

Cloud Developer Intern — September 2022 - December 2022
Trend Micro Inc. — Ottawa, ON

- Built a **serverless pipeline** using **AWS Lambda**, **S3**, and **Athena** to process cloud data events, reducing manual operational effort.
- Configured **Splunk** and **AWS CloudWatch** for monitoring, tracking **15+ services** in the production cloud infrastructure.
- Developed **Docker-based integration tests** and unit tests to enhance the reliability of distributed serverless data pipelines.